

Tide retrospective:
*anharmonic-partition-deriv (partial: n=1 integrability witness
added)*

Claude (Opus 4.7, 1M context)

2026-05-23

Abstract

Partial progress on closing the `partition_hasDerivAt` sorry from `AnharmonicGibbsRegularity.lean`. Added the public `integrable_x_mul_exp_neg_t_anharmonic` lemma — the $n = 1$ mirror of `integrable_exp_neg_t_anharmonic`, providing integrability of $x \cdot \exp(-t \cdot L_{\text{anh}}(x))$ on $\mathbb{R}$ under the discriminant condition. This is the integrability witness for the RHS of `partition_hasDerivAt`'s expected derivative ($\int (-t \cdot x) \cdot \exp(-t \cdot L_{\text{anh}}(x)) dx$). The full `partition_hasDerivAt` proof itself (via either MGF route or direct dominated convergence) remains as a sorry; this tide lands one of the required lemmas.

1 What landed

One new theorem in `Laplace/OneD/AnharmonicGibbsRegularity.lean`:

- `integrable_x_mul_exp_neg_t_anharmonic` (~ 35 lines). Public witness for $x \cdot \exp(-t \cdot L_{\text{anh}}(x))$ integrable on $\mathbb{R}$ under $\alpha^2 < 3\lambda\gamma, \lambda > 0, \gamma > 0, t > 0$. Proof by Gaussian domination via `anharmonic_coercive` and `integrable_abs_pow_mul_exp_neg_mul_sq` at $n = 1$. Mirrors the existing $n = 0$ case essentially line-for-line.

Build status: clean (only the existing sorry warning on `partition_hasDerivAt`).

2 Why stop here

The `partition_hasDerivAt` proof itself was attempted but abandoned mid-write per the laplace CLAUDE.md thrashing rule. The structure is clear:

1. Apply `MeasureTheory.hasDerivAt_integral_of_dominated_loc_of_deriv_le`.
2. Need a bound function $\text{bound}(x) := |t \cdot x| \cdot \exp(-t \cdot L_{\text{anh}}(x)) \cdot \exp(t \cdot |x|)$. Integrability follows from coercivity + AM-GM: $|x| \leq (c/2)x^2 + 1/(2c)$ gives $-tc \cdot x^2 + t|x| \leq -(tc/2)x^2 + t/(2c)$, so $\text{bound}(x) \leq |t \cdot x| \cdot \exp(t/(2c)) \cdot \exp(-(tc/2)x^2)$, Gaussian-integrable via the $n = 1$ Mathlib lemma.
3. Pointwise derivative via chain rule of `HasDerivAt.exp`, `HasDerivAt.const_add`, etc.
4. Discharge AE-measurability conditions via continuity.

Estimated ~ 80 – 120 lines of bounded but technically involved Lean. Without an MCP Lean LSP for iterative goal-state inspection, attempting this in one shot via `lake build` is high-risk thrashing. The partial commit here lands one of the four pieces; the remaining three are still pending.

3 Pipeline lessons

- **Partial-progress commits are valuable.** Even one integrability lemma is reusable: the next session attempting the full proof can call `integrable_x_mul_exp_neg_t_anharmonic` directly without re-deriving.
- **Mirror the existing $n = 0$ pattern when adding $n = k$.** The integrability proofs in this file follow a strict template: `anharmonic_coercive` $\Rightarrow$ Gaussian dominator $\Rightarrow$ `Integrable.mono` $\Rightarrow$ pointwise inequality via `Real.exp_le_exp.mpr` + `linarith`. Future $n = 2, 3, \dots$ instances should be 30-line copy-edits.

4 Follow-up tide

Close `partition_hasDerivAt`: ~ 80 – 120 lines, sharply scoped, all prerequisite integrability lemmas now public. The next session can apply the MGF route (per addendum 3 of `projects/primer/tide-log/2026-05-2`) or the direct DCT route — both are documented in the docstring at the sorry site.

After that closes: 6 `GibbsObservable` instances (30-50 lines each) and the headline corollary `cov_h_id_id_deriv_anharmonic_eq_neg_t_kappa3` (~ 20 lines).